

Feature 3 (Deep Mode/Cram Mode):

Problem: A single study guide cannot effectively serve students with diverse priorities. Students who have more time and want to go in depth on the material cannot do so, and students who have little time and need to focus on important material must sift through the study guide to identify key topics.

Target Users: STEM students who want to learn the material with different time constraints.

Proposed Solution: If students are given the option to choose between Deep mode and Cram mode, with Deep mode going in depth on the material and Cram mode focusing only on important topics.

Goals:

- **Primary metric:**
 - **Deep Mode:** In the feedback provided after the study guide, the student can indicate whether the guide is comprehensive enough. The number of students who think the guide is not comprehensive enough should be fewer than 10%.
 - **Cram Mode:** In the feedback form provided after the study guide, the student can indicate whether any material is missing/repeated. This percentage needs to be below 10%.
 - **Shared Metric:** The quiz score; directional signal only, not attributable to mode choice given confounds (prior knowledge, guessing, topic-skipping)
- **Guardrails:**
 - The survey's topic-missing rate doesn't go above 10%.

Scope of the solution:

Out of scope:

MVP / Functional Requirements

Core flow:

- [P0] After indexing materials. Students can choose between Deep mode and cram mode
- [P0] If Deep mode is selected, the system retrieves all the material that is indexed and generates an in-depth study guide.
- [P0] If Cram mode is selected, the system retrieves the bolded material from the indexed material and generates a study guide focusing only on important topics.

Edge case handling:

Cram Mode

- [P0] If the study material doesn't have any bolded text or headings, the students get a choice to upload any material list given by the professor.
- [P0] If the student doesn't have an important material list, then the AI gets to decide the important topic based on the material given, the frequency of things, fundamentals, etc.
- [P0] If the study guide is not generated, an error will be displayed, and the student will be asked to retry. If the study guide can still not be generated, the admin will be notified.

Feedback/telemetry:

- [P0] After the study guide generation, the student will be asked how useful the study guide was and get to fill out a small survey where they will be able to indicate if the topics were missing or if topics are repeating. The feedback will be given to the admin for fixing.